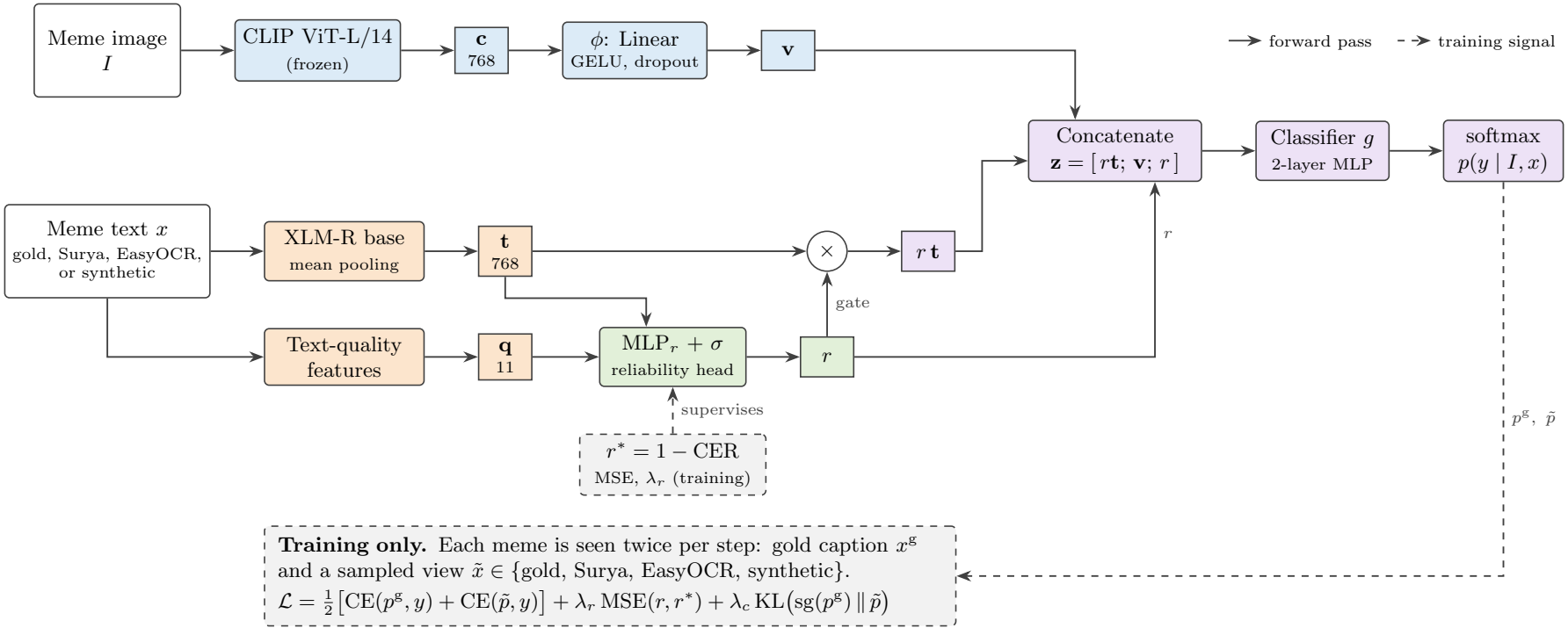

Meme image I
CLIP ViT-L/14 (frozen)
c 768
ϕ: Linear GELU, dropout
v
forward pass
training signal
Concatenate $\mathbf{z} = [\, r\mathbf{t};\, \mathbf{v};\, r \,]$
Classifier g 2-layer MLP
softmax $p(y \mid I, x)$
Meme text x gold, Surya, EasyOCR, or synthetic
XLM-R base mean pooling
t 768
$\times$
$r\,\mathbf{t}$
gate
r
Text-quality features
q 11
$\mathrm{MLP}_r + \sigma$ reliability head
r
supervises
$r^* = 1 - \mathrm{CER}$ MSE, λ_r (training)
$p^{\mathrm{g}},\ \tilde{p}$
Training only. Each meme is seen twice per step: gold caption x^{g} and a sampled view $\tilde{x} \in$ {gold, Surya, EasyOCR, synthetic}.
$\mathcal{L} = \frac{1}{2}\big[\mathrm{CE}(p^{\mathrm{g}}, y) + \mathrm{CE}(\tilde{p}, y)\big] + \lambda_r\,\mathrm{MSE}(r, r^*) + \lambda_c\,\mathrm{KL}\big(\mathrm{sg}(p^{\mathrm{g}}) \,\|\, \tilde{p}\big)$